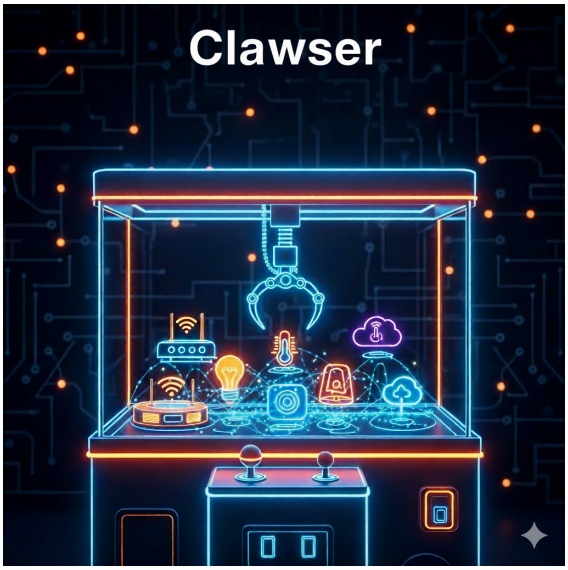
Department of Computer Science & Engineering
The University of Texas at Arlington

System Requirements Specification
CSE 4316: Senior Design I
Fall 2025



Clawser
IOT Claw Game

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1 Product Concept

Our product is an IoT-enabled claw machine system that integrates a web mobile application, token management, and secure access features to deliver an interactive remote-control experience. The system combines hardware (the claw machine and control board) with cloud-based software to support user authentication, game control, payment simulation, and administrative monitoring. The overall goal is to design a scalable and secure platform suitable for both entertainment and commercial environments.

1.1 Purpose and Use

The primary purpose of the system is to allow users to remotely control a physical claw machine through mobile interface, providing a real-time gaming experience. Users will sign in via a single sign-on (Google) mechanism, purchase or simulate tokens through a secure token system, and interact with the machine using intuitive controls. Administrators can access a diagnostic dashboard to monitor active machines, detect faults, and disable malfunctioning units. The system emphasizes standard secure practices, user accessibility, and responsive operation.

1.2 Intended Audience

The system is designed for both casual players and arcade operators. Casual users will enjoy the convenience of playing a physical claw machine remotely, while business operators can benefit from a scalable, cloud-based platform capable of supporting multiple machines and real-time monitoring. Future expansion could allow broader commercial applications such as remote vending or entertainment network integration.

The intended audience includes:

- End users seeking an engaging, online-connected arcade experience.
- Business owners or arcade operators who can deploy the system to expand customer engagement through remote play options.
- Developers and administrators responsible for maintaining machine uptime, verifying transactions, and managing multiple units.

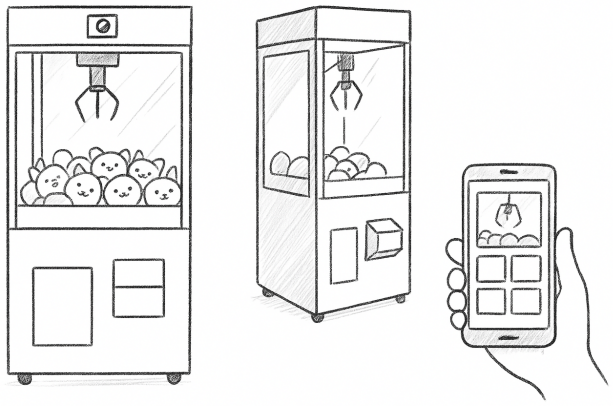

Figure 1: IoT Claw Machine conceptual drawing

2 Product Description

This section provides an overview of the IoT Claw Game System, describing its major components, functions, and operational flow. The system integrates both hardware and software to allow users to remotely control a physical claw machine through a secure web or mobile interface. Its primary operational aspects include user authentication via Google single sign-on, a virtual token management system, and real-time communication between the client application and the claw machine hardware. The following subsections detail the system's features and functions, external inputs and outputs, and its operational interfaces.

2.1 Features & Functions

The IoT Claw Game System enables interactive remote gameplay by linking a physical claw machine to an online control platform. The key features and functions include:

- **Remote Claw Operation:** Players can manipulate the claw in real time using on-screen controls that send directional commands to the machines motors. The control interface uses low-latency communication through WebSockets or MQTT protocols to maintain precise movement response.
- **Token Management:** A secure token system simulates the arcade payment process. Tokens are deducted for each play session and updated in the users account balance. This allows both simulated and future real-payment integration.
- **Single Sign-On (SSO):** Users sign in securely with their Google accounts, eliminating the need for local password storage and increasing accessibility and data security.
- **Multi-Camera Live Stream:** The system supports multiple camera angles such as top and side views to give players full visibility of the claws position and prize area. The streaming is handled via OBS and WebRTC to ensure smooth and synchronized video delivery.
- **Diagnostic and Fault Detection System:** Built-in diagnostic tools provide feedback on hardware performance, sensor activity, and potential communication faults, allowing maintainers to identify issues proactively.
- **Scalability and Expandability:** The modular architecture allows additional machines or devices to be integrated easily through the backend infrastructure, supporting future deployment in arcade venues or entertainment centers.
- **Remote Operation:** Users can control the physical claw through on-screen joystick or button inputs that transmit real-time commands to the machines control board.
- **Token Simulation System:** Each play requires virtual tokens that are purchased or simulated through the platform, reflecting real-world arcade mechanics while enabling flexible credit management.
- **Secure Authentication:** Google single sign-on (SSO) ensures that only verified users can access the system, enhancing both usability and data protection.
- **Real-Time Feedback:** The system provides continuous video streaming and machine status updates, allowing players to see claw movement, dropped items, or mechanical errors as they occur.
- **Scalability:** The design supports future expansion to multiple machines or arcade environments, minimizing reconfiguration time and ensuring consistent integration across deployments.

Primary Components

- **Claw Cabinet Assembly:** Includes motors, gantry arm, sensors, LEDs, and camera modules attached to an aluminum extrusion frame.
- **Local Computer Controller:** Acts as the local hardware interface that translates network commands into motor movements and sensor readings.
- **Backend Server:** Handles real-time communication, user authentication, and video streaming services using Google Cloud.
- **Frontend Application:** A React Native-based interface for user control, token management, and live video display.

External Elements

- **Internet / Google Cloud Platform:** Provides hosting, streaming, and authentication services.
- **User Devices (Mobile):** Send inputs and receive video feedback.
- **OBS Server:** Manages video encoding and stream relay to connected clients.

Together, these functions establish a seamless connection between digital commands and physical hardware, creating an engaging remote arcade experience that can be adapted for educational, entertainment, or commercial use.

2.2 External Inputs & Outputs

The IoT Claw Game System relies on several key data exchanges between users, administrators, and the hardware subsystem.

External Inputs:

- User authentication data from Google SSO.
- Token purchase or simulation requests.
- Control signals from the user interface (joystick, directional buttons, drop command).
- Feeds from the cameras attached to the claw machine (front, side, and top view).
- Sensor feedback indicating prize detection.

External Outputs:

- Real-time machine video feed and motion feedback transmitted to the user's interface.
- Token balance updates and play results (win/loss notifications).
- Status indicators showing machine availability, connection health, and system alerts.

All input and output interactions are transmitted through secure communication channels, ensuring data integrity and consistent performance across networked environments.

2.3 Product Interfaces

The product features distinct interfaces tailored for end users and maintainers:

- **End-User Interface:** A responsive mobile or web application that presents a live video feed of the claw machine, control buttons for directional movement, and token indicators. The interface emphasizes simplicity and accessibility, allowing players to intuitively control gameplay with minimal delay.
- **Maintenance Interface:** A basic control and diagnostic screen accessible locally at the machine, enabling technicians to calibrate movement systems, verify connectivity, and perform repairs.

These interfaces provide a cohesive operational environment that supports end-user engagement, system management, and long-term maintainability. Both interfaces reflects the same design philosophy clear visualization, security, and scalability to ensure the products adaptability for future development and commercial deployment.

3 Customer Requirements

The IoT Claw Game is designed to be an engaging, interactive, and secure experience for users who use the claw machine remotely through a web or mobile interface. The following customer requirements define the "look and feel" of the product, as expected by the sponsor and intended end users. They outline how the application should appear and respond, including its layout, accessibility, gameplay interface, live video integration, and security features. Each requirement described in this section reflects features that the customer will directly see or interact with during normal operation of the product.

3.1 Gameplay Control Interface

3.1.1 Description

The system shall provide a gameplay control interface with virtual buttons for directional movement (left, right, up, down) and a Grab button for claw operation. Controls shall be arranged in a cross-pattern at the bottom of the screen. Buttons shall respond to press-and-hold input, moving the claw only while pressed and stopping movement immediately upon release.

3.1.2 Source

Sponsor

3.1.3 Constraints

Controls are limited to one active session per claw machine.

3.1.4 Standards

ISO 9421-112, WCAG 2.1 AA

3.1.5 Priority

Critical

3.2 Live Video Display

3.2.1 Description

The system shall display live video feeds from three cameras (front, side, and top views) in the gameplay interface, with a toggle button to switch between views.

3.2.2 Source

Sponsor

3.2.3 Constraints

The website should allow for the dynamic change of video feeds without a full website reload. Video streams must have minimal delay to preserve real-time control perception.

3.2.4 Standards

ISO/IEC 23009-1, ISO 9241-303

3.2.5 Priority

Critical

3.3 User Experience and Accessibility

3.3.1 Description

The system shall provide an intuitive and accessible interface designed for ease of use. All primary actions ranging from login, token use, and gameplay shall be clearly visible within the main interface. Accessibility features such as readable fonts and high-contrast themes shall be included to ensure usability for a wide range of users.

3.3.2 Source

Sponsor

3.3.3 Constraints

All interactive elements shall be operable without requiring audio or color perception.

3.3.4 Standards

WCAG 2.1 AA (as according to UTA policy), ISO 9421-110

3.3.5 Priority

High

3.4 Security and Login Interface

3.4.1 Description

The system shall include a login screen that supports Google Single Sign-On (SSO) with a Sign in with Google button centered on the page. After authentication, the system shall display a clear confirmation message indicating successful login and secure connection.

3.4.2 Source

Sponsor

3.4.3 Constraints

Must comply with Google OAuth 2.0 and platform usage policies.

3.4.4 Standards

IETF RFC 6749 (OAuth 2.0 Authorization Framework), TLS 1.3

3.4.5 Priority

High

3.5 Token Purchase Simulation

3.5.1 Description

The system shall include a token management interface that allows users to simulate purchasing tokens via a Buy Tokens button. The user current token balance shall be displayed in the header.

3.5.2 Source

Sponsor

3.5.3 Constraints

Tokens are virtual and non-transferable. Token balances must persist for the duration of a session.

3.5.4 Standards

ISO 9421-112, OWASP ASVS

3.5.5 Priority

Medium

4 Packaging Requirements

The IoT Claw Machine will be delivered as a fully assembled and functional unit with all required hardware, wiring, and pre-installed software. The below packaging requirements specify how the physical cabinet, electronics, and accompanying materials are bundled and delivered to ensure safety, ease of setup, and professional appearance. The packaging must also protect sensitive IoT components (controllers, sensors, cameras) during transit and comply with applicable electrical safety and material handling standards.

4.1 Enclosure and Cabinet packaging

4.1.1 Description

The IoT Claw Machine cabinet will be enclosed in an aluminum and acrylic frame with plates and brackets.

4.1.2 Source

Customer

4.1.3 Constraints

Maintain structural integrity during transport and assembly, the design should also minimize sharp edges and pinch points. The frame must conform to Misumi part standards.

4.1.4 Standards

UL 22 (Amusment and Gaming Machines)

4.1.5 Priority

Critical

4.2 Software Delivery and Setup

4.2.1 Description

The IoT Claw Machine software stack will be preloaded on the onboard computer and cloud instance. The software stack shall consist of controller firmware, server application, and web interface. Upon power up, the system will automatically start the required services and establish IoT connectivity.

4.2.2 Source

Customer

4.2.3 Constraints

Software must run on a Single Board Computer or a Desktop Computer. All credentials and API keys must be stored securely prior to delivery. A stable Internet connection via Ethernet (If possible with WiFi fallback).

4.2.4 Standards

IEEE, ACM, ISO

4.2.5 Priority

Critical

4.3 Documentation and labeling

4.3.1 Description

A printed user guide and safety sheet will be included. Labels identifying major components, serial numbers, and voltage ratings shall be affixed inside the cabinet near their respective parts.

4.3.2 Source

Customer

4.3.3 Constraints

Must be clearly legible and durable against wear and tear. Labeling text and symbols must comply with safety labeling standards. Documentation to include troubleshooting and maintenance sections.

4.3.4 Standards

IEEE, ISO (9001-quality management, 27001-information security, 28219 - general product marking)

4.3.5 Priority

Low

4.4 Power Cable Management

4.4.1 Description

The IoT Claw Machine will include neatly routed wiring with labeled power cables. Power supply units will be mounted internally with fuses in compliance with low voltage safety standards. All user facing cables will meet cable and wiring saftey standards.

4.4.2 Source

Customer

4.4.3 Constraints

Total load must not exceed rated power for the supplied PSU. Cable routing must prevent pinching and exposure points.

4.4.4 Standards

National Electrical Code (NEC), International standard for the electrical equipment of machines (IEC 60204-1)

4.4.5 Priority

High

5 Performance Requirements

Our project is inspired by other applications. After extensive research into what makes these applications successful, as well as meetings with our sponsor to determine key requirements, we have the following on our list of the most important. We believe it is essential for the user to be able to control the claw machine in real-time through the app, as well as have the ability to switch between different camera angles of the claw machine. The user should also be able to create an account on our application, and the application should have the ability to keep track of prizes won by the user. The claw machine must have sensors to be able to detect when a prize has been won and communicate it to the cloud. Admins of the application will be able to control the strength of the claw.

5.1 Real-Time Control

5.1.1 Description

The user must be able to control the claw machine in real-time through the app. Ensuring that pressing the buttons gives the illusion of controlling the live movement of the claw in the user's POV is a crucial part of our project.

5.1.2 Source

Sponsor

5.1.3 Constraints

The system must be capable of maintaining responsive control and video communication over normal internet speeds. The claw machine will be located at UTA, so it will depend on the UTA network. The claw machine communication will depend on the specifications of the selected microcontroller, motor drivers, and cameras that we will purchase for the project.

5.1.4 Standards

IEEE (802.11/Wi-Fi Alliance Certification - IoT Protocol Standards), ISO 9241-210 (Human-Centred Design for Interactive Systems)

5.1.5 Priority

Critical

5.2 Live video stream

5.2.1 Description

The user must be able to switch and view different angles of the claw machine. The claw machine will have an angle from inside the claw machine as well as outside it with one camera angle being the bare minimum for functionality of our project. The video stream also needs to be fast enough to sync with the latency of the claw machine controls to give the appearance of live control.

5.2.2 Source

Sponsor

5.2.3 Constraints

We will be using the UTA network to stream, so we are limited to the speeds available through the network. The video stream must be compatible to be streamed using the API in our application. The exact frame rate, resolution, and protocol will depend on the selected software and camera hardware once additional parts are purchased and further testing begins.

5.2.4 Standards

ISO 9241-210 (Human-Centred Design for Interactive Systems), IEEE (IoT Protocol Standards)

5.2.5 Priority

Critical

5.3 Account creation

5.3.1 Description

The user must be able to log in and have an account on the application. This will be done through a stored database or utilizing third party authenticators to create user accounts. These accounts need to have the ability to keep track of game tokens, prizes and other information necessary.

5.3.2 Source

Sponsor

5.3.3 Constraints

We must meet industry standards, using either a managed identity provider (e.g. OAuth2) or email and password with secure password storage. Either method we choose must be compatible with the cloud framework.

5.3.4 Standards

IETF RFC 6749 (OAuth 2.0 Authorization Framework), TLS 1.3

5.3.5 Priority

High

5.4 Game start-up speed

5.4.1 Description

Once the user clicks on a game to play, the app should bring up the video stream and controls in a reasonable amount of time to keep users engaged. Neither the video nor the controls should have drastically different load up speeds than the other.

5.4.2 Source

Sponsor

5.4.3 Constraints

The methods and protocols we use to stream the controls and video over to the user is going to determine the speeds we are able to achieve. The internet speeds of the UTA network will play a major factor in our load up speeds.

5.4.4 Standards

IEC 61499, ISO 9241-210

5.4.5 Priority

High

5.5 Prize detection

5.5.1 Description

The claw machine must have sensors to detect when a prize has been won and be able to communicate that to the application. The message should be sent by the sensors fast enough to be received by the user shortly after their prize drops through the chute. The sensors must have the ability to perform under low lighting conditions.

5.5.2 Source

Sponsor

5.5.3 Constraints

The sensors we purchase must be mountable and be able to fit in the chute area of the claw machine once the machine is built. Sensors must operate under variable lighting, reflections, and dust expected in arcade conditions, with false positives and detection errors kept to a minimum. The computer with the claw machine needs to be able to communicate with the cloud to update the user's point of view and alert them that the machine detected a prize was won.

5.5.4 Standards

IEEE (IoT Protocol Standards)

5.5.5 Priority

High

5.6 Claw strength control

5.6.1 Description

The admins of the application must be able to control the strength of the claw. This means that admins need to have the ability to change user settings on the claw machine to change difficulty when desired to by the admins.

5.6.2 Source

Sponsor

5.6.3 Constraints

The claw must have various degrees of strength to change the diffulty to obtain a prize, however, it must not damage the prize.

5.6.4 Standards

IEC, IEEE

5.6.5 Priority

Medium

6 Safety Requirements

The IoT Claw Machine will adhere to the safety requirements of the laboratory used to build the machine as well as follow any requirements from our sponsor. Engineering safety requirements for the cabinet will be followed to ensure no harm comes to the equipment or the administrators. Any equipment used will be done with the approval of the laboratory and returned once finished with OSHA and LOTO procedures in mind. Electrical, fire, and data safety are considered for hazard protection and security.

6.1 Laboratory equipment lockout/tagout (LOTO) procedures

6.1.1 Description

Any fabrication equipment provided used in the development of the project shall be used in accordance with OSHA standard LOTO procedures. Locks and tags are installed on all equipment items that present use hazards, and ONLY the course instructor or designated teaching assistants may remove a lock. All locks will be immediately replaced once the equipment is no longer in use.

6.1.2 Source

CSE Senior Design laboratory policy

6.1.3 Constraints

Equipment usage, due to lock removal policies, will be limited to availability of the course instructor and designed teaching assistants.

6.1.4 Standards

Occupational Safety and Health Standards 1910.147 - The control of hazardous energy (lockout/tagout).

6.1.5 Priority

Critical

6.2 National Electric Code (NEC) wiring compliance

6.2.1 Description

Any electrical wiring must be completed in compliance with all requirements specified in the National Electric Code. This includes wire runs, insulation, grounding, enclosures, over-current protection, and all other specifications.

6.2.2 Source

CSE Senior Design laboratory policy

6.2.3 Constraints

High voltage power sources, as defined in NFPA 70, will be avoided as much as possible in order to minimize potential hazards.

6.2.4 Standards

NFPA 70

6.2.5 Priority

Critical

6.3 RIA robotic manipulator safety standards

6.3.1 Description

Robotic manipulators, if used, will either housed in a compliant lockout cell with all required safety interlocks, or certified as a "collaborative" unit from the manufacturer.

6.3.2 Source

CSE Senior Design laboratory policy

6.3.3 Constraints

Collaborative robotic manipulators will be preferred over non-collaborative units in order to minimize potential hazards. Sourcing and use of any required safety interlock mechanisms will be the responsibility of the engineering team.

6.3.4 Standards

ANSI/RIA R15.06-2012 American National Standard for Industrial Robots and Robot Systems, RIA TR15.606-2016 Collaborative Robots

6.3.5 Priority

Critical

6.4 Enclosure and User Electrical Safety

6.4.1 Description

The IoT Claw Machine enclosure will be designed to prevent user contact with any internal wiring or high-current components. All external interfaces (buttons, cables, and connectors) must be rated for low-voltage operation and enclosed to prevent electric shock. Any exposed metal parts will be properly grounded.

6.4.2 Source

CSE Senior Design laboratory policy, additional sources

6.4.3 Constraints

All high-voltage wiring will remain fully enclosed within the cabinet. Access panels will require tools to open. All power distribution within the cabinet must operate below 48V DC where feasible to maintain user safety.

6.4.4 Standards

IEC 60204-1, NEC, UL 62368-1, EN 55032

6.4.5 Priority

Critical

6.5 Emergency Stop and Power Isolation

6.5.1 Description

The IoT Claw Machine shall include a clearly marked emergency stop (E-Stop) button that immediately cuts power to the motors and actuators when pressed. The E-Stop must be accessible from the front of the cabinet and resettable only by authorized personnel.

6.5.2 Source

CSE Senior Design laboratory policy

6.5.3 Constraints

The emergency stop mechanism must fully disconnect the actuator control circuits within 1 second.

6.5.4 Standards

NEC, UL 62368-1

6.5.5 Priority

Medium

6.6 Thermal and Fire Safety

6.6.1 Description

The IoT Claw Machine must be designed to minimize overheating risks. All electronic components will include adequate ventilation and thermal management (fans or heat sinks). Flammable materials will be avoided near power circuits.

6.6.2 Source

CSE laboratory safety policy

6.6.3 Constraints

Internal temperature must remain below 45C during continuous operation.

6.6.4 Standards

UL 62368-1

6.6.5 Priority

High

6.7 IoT Network and Data Safety

6.7.1 Description

All critical IoT system data including session logs, gantry position, and other user configuration settings shall be automatically backed up to a secure cloud storage to ensure data integrity and easy recovery in the event of a system failure, data corruption, or network interruption.

6.7.2 Source

Customer requirements

6.7.3 Constraints

Backup data must be encrypted and verified for integrity after each backup. Backup and restore operations must be done in a way that does not interrupt gameplay. or video streaming.

6.7.4 Standards

ISO 27001: Information Security Management Systems
ISO 27040: Storage Security (Can use both for Backups)

6.7.5 Priority

High

6.8 Moving Mechanism and Pinch Hazard Prevention

6.8.1 Description

The claw gantry system will be designed to prevent user contact with moving parts during operation. The access door will include an interlock to disable motion when open. Mechanism travel limits will be implemented in software and hardware to avoid overextension.

6.8.2 Source

Sponsor

6.8.3 Constraints

All moving axes must stop automatically if a limit switch is triggered or if mechanical interference occurs.

6.8.4 Standards

UL-22

6.8.5 Priority

Critical

7 Security Requirements

Ensuring the IoT Claw Machine provides a secure and trustworthy experience for users is critical for this system. It is essential that users can log in securely through a trusted provider and that their personal information is kept safe and private. In addition, the connection between the application and physical machine must be secure to prevent unauthorized interference. The system must also enforce user permission levels (players vs administrators) and prevent player access to administrator controls. The following section lists the specific requirements to implement these security measures as well others.

7.1 Authentication and Authorization

7.1.1 Description

The system shall provide secure account login using Google Single Sign-On. Users must login to access the application. The system must also support role-based access control for players and administrators. Administrators shall have elevated permissions such as viewing diagnostic information, activating or deactivating machines, etc.

7.1.2 Source

Sponsor

7.1.3 Constraints

Compatability with our selected cloud framework.

7.1.4 Standards

IETF RFC 6749 (OAuth 2.0 Authorization Framework), OWASP ASVS

7.1.5 Priority

Critical

7.2 Data Protection

7.2.1 Description

The system shall securely store all personally identifiable information collected from the identity provider (Google Single Sign-On) such as name and email. This also applies to user-specific data such as prize history and token balance.

7.2.2 Source

Sponsor

7.2.3 Constraints

Database solution must support modern encryption standards.

7.2.4 Standards

ISO 27018, OWASP ASVS

7.2.5 Priority

High

7.3 Secure Communication

7.3.1 Description

All system communications, such as user control commands, live video feed, and prize detection, shall be transmitted over secure channels to ensure game integrity and prevent unauthorized interference.

7.3.2 Source

Sponsor

7.3.3 Constraints

Communication devices must support encryption protocols (TLS/SSL).

7.3.4 Standards

OWASP ASVS

7.3.5 Priority

Critical

7.4 Session Integrity

7.4.1 Description

Once a user has been authenticated, the system shall create and manage a secure session. All actions, such as sending commands to the claw, must be tied to a valid, active user session. Sessions shall expire after a period of inactivity.

7.4.2 Source

Sponsor

7.4.3 Constraints

Session states must be securely managed on the server to prevent tampering.

7.4.4 Standards

OWASP ASVS

7.4.5 Priority

High

7.5 Logging

7.5.1 Description

The system shall maintain a time-stamped log of all security relevant events. This inculdes, user authentications, administrator actions, user sessions, and other similar events. The logs must be viewable to users with administative privileges.

7.5.2 Source

Sponsor

7.5.3 Constraints

Storage space for log retention. Protect logs against unauthorized modification or deletion.

7.5.4 Standards

OWASP ASVS

7.5.5 Priority

High

8 Maintenance & Support Requirements

The claw machine will allow for administrators to receive error reports through a simple dashboard from our cloud services. For maintainers, there will be a manual to troubleshoot hardware errors such as claw strength or replacing the claw gantry. To ensure the best results, the machine should have routine checks every week.

8.1 Troubleshooting Manual

8.1.1 Description

A manual will be provided for the maintainer team to review in case of any errors with the claw. The manual shall include steps to remove or replace hardware, the tools recommended to use for the hardware, and how to detect which aspect of the IoT claw machine is failing (through software or hardware). The troubleshooting manual shall include warnings and procedures for handling the claw machine.

8.1.2 Source

Team

8.1.3 Constraints

The manual should be clear to read with diagrams and labels when needed. The manual should include safety and security information to prevent injury and recommendations on handling electronics.

8.1.4 Standards

ISO, IEC, IEEE

8.1.5 Priority

Medium

8.2 Simple Administrator Dashboard

8.2.1 Description

The application will provide a simple dashboard for the administrators to use. The dashboard will show the claw machine connected to it. A report will be sent through the cloud from the machine and recieved on the dashboard.

8.2.2 Source

Team

8.2.3 Constraints

There should be minimal delay/latency when receiving error messages from the claw machine.

8.2.4 Standards

IEEE, IEC

8.2.5 Priority

Medium

8.3 Routine Checks

8.3.1 Description

The maintenance team should conduct routine checks on the machine and application. These checks should include the cameras, the latency, the claw gantry reactivity and strength, as well as any software related errors. This includes any exterior factors that may influence gameplay such as lighting and maintaining cleaness around and inside the machine such as lubricating moving parts and checking for any electrical systems.

8.3.2 Source

Team

8.3.3 Constraints

To complete a full routine check, at least one team member will need to be with the claw machine.

8.3.4 Standards

IEC, UL

8.3.5 Priority

Medium

9 Other Requirements

The claw machine and application are completed with scalablity in mind. Keeping the cabinet light to move it around easier and within a reasonable price so more can be built if desired. The application will allow for updates to be made to increase the machines and handle more users playing and viewing our online arcade. The application and machine will be made so that each part of the project can work at the same time with low latency.

9.1 Scalability

9.1.1 Description

As per our customers' needs, the IoT claw machine will be capable of expanding for more than one machine. The application will fulfill load and stress tests for a many users to play different claw machines or view the streams at once.

9.1.2 Source

Customer

9.1.3 Constraints

The interoperability of the claw machine is important to allow multiple users to interact with the application and see the livestream. The claw machine should only respond to the person currently playing and not to someone in the queue watching the stream.

9.1.4 Standards

UL, NES, EN, IEC, IEEE, ACM

9.1.5 Priority

Medium

10 Future Items

The following items are requirements to consider for expanding the IoT claw machine but will not be fulfilled. Per our sponsor's desire, the project is designed with scalability in mind such as handling multiple cabinets and allowing administrators to monitor activity through an advanced dashboard. To make the project more accessible, the application should support more than one language but time constraints prevent it from being added in the prototype.

10.1 Multiple Cabinent/Game Support

10.1.1 Description

Allow for the deployment of multiple cabinents with various games and functions to the server and allow users to view and interact with them.

10.1.2 Source

Sponsor

10.1.3 Constraints

Allow machines to create streams, and be controlled by the server. Users can view different games connected to the server.Once connected, users can play and control the arbitrary game it represents. Security must be emphasized as to allow only trusted machines to publish to the platform.

10.1.4 Standards

Websockets (RFC6455), WebRTC (RFC8835)

10.1.5 Priority

Medium

10.2 Multi-Language Support

10.2.1 Description

Integrate multiple language options to increase the potential pool of users that would be able to use the machine.

10.2.2 Source

Team

10.2.3 Constraints

Storing and switching between language files should not increase latency. Additionally, translations must be accurate and maintain UI layout consistancy. Only a small number of languages can be added for time.

10.2.4 Standards

ISO 17100, WCAG 2.1 AA

10.2.5 Priority

Low

10.3 Advanced Admin Analytics Dashboard

10.3.1 Description

To fascilitate the addition of multiple cabinets/game support as well measure and monitor user activity on the app, additional charts, usage stats, and player analytics will need to be added in an admin panel.

10.3.2 Source

Team

10.3.3 Constraints

User data privacy must be maintained. Additionally, capturing analytics should not increase latency. Dashboard analytics will need to be refreshed at a modest (perhaps hourly) rate to keep an up to date look at the performace of the system.

10.3.4 Standards

OWASP, ISO 27001

10.3.5 Priority

Medium

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